

Compare data distributions



- 1
- a Paris
 - b Rome
 - c New York
 - d Tokyo and Rome
 - e 0%
 - f 20%
- 2
- a 24
 - b 10
 - c 67
 - d 119
 - e 20.17%
 - f 13.4
 - g 23.8
 - h Pens, because the daily mean is higher.
- 3 Humbleton, because the mean is higher than the median, so most house prices are higher than \$950 000.
- 4
- a 17
 - b 16
 - c 29.5
 - d 37.5
 - e No, even though their ranges are similar, their means are too far apart.
- 5
- a Positively skewed
 - b Berger's Burgers
 - c Fry's Fries, because the range is smaller.
 - d Fry's Fries, because it has a higher median.
- 6
- a 5.2
 - b 8.9
 - c 3
 - d 3
 - e No. The results are mostly 4 or 5 out of 10 and the mean is only 5.2.
 - f Yes. The results are mostly 9 and 10 out of 10, and the mean is 8.9.

7



- a
- i 2.6 ii 3 iii 3 iv 4
- b
- i 8.1 ii 8 iii 8 iv 4
- c Class 2, because their mean, median and mode were higher.

8

- a
- i 7.6 ii 7 iii 7 iv 5
- b
- i 7.4 ii 7.5 iii 7 iv 1
- c Person A, because they have the higher range.
- d Person A, because their mean and mode is higher.

9

- a
- i \$77 142.86 ii \$80 000 iii \$80 000 iv \$10 000
- b
- i \$72 857.14 ii \$70 000 iii \$70 000 iv \$10 000
- c The men, because they have the higher mean, median and mode.

10

- a Lower b 0
- c 2 d Positively skewed
- e 4 f Block B
- g No

11

- a Perth b Melbourne c Perth

12

- a United States b Sweden c Perceived

13

- a NSW b D c November d 15 270.84 units

14

- a O+ b AB− c B+, AB d 12%
- e 39 657 759

15



- a Section 1
- b Section 2
- c They have the same range.
- d The mean is equal to the median.
- e Section 2

16

- a Female
- b Male
- c Female
- d Female
- e Male

17

- a Positively
- b 30 – 34
- c 30 – 34
- d Female

Parallel box plots

18

- a Tobias
- b Ned
- c Ned
- d \$5000

19

- a Red bricks have a higher median and higher maximum value.
- b Yellow bricks have a higher minimum and a smaller range so their strength is more consistent.

20

a

	Manufacturer A	Manufacturer B
Median	4000	5000
Lower Quartile	2500	3500
Upper Quartile	4500	6000
Range	4000	6500
Interquartile Range	2000	2500

- b Manufacturer B, because 50% of their light bulbs last longer than all of Manufacturer A's light bulbs.

21



a

i 13

ii 10

iii 18

iv 13

v 8

b

i 13

ii 11

iii 14

iv 11

v 3

- c Class 9P. The top 25% of their marks are between 18 and 20, while the top 25% for 9Q is between 14 and 20.

22

- a Weightlifter A, because they have a smaller range and interquartile range.
- b Weightlifter B, because their upper quartile and maximum number of repetitions are both higher than Weightlifter A's maximum.

23

a

	English	Mathematics
Median	10	10
Lower Quartile	8	7
Upper Quartile	12	15
Range	8	15
Interquartile Range	4	8

- b Mathematics. More than 25% of the mathematics marks are above all of the English marks.

24

a 49

b 50

c 12

d Carl

e Carl

25

a 12 bpm

b 12 bpm

c 12 bpm

d 52 bpm

e 56 bpm

f Before

g 16 bpm

26

a Bookstore A

b \$10

c True

d True

27

a 0.5

b 0.75

c 3rd

d 2010

28



- a Film C, because all the ages are 18 or older.
- b Film B, because the median age is 16.
- c Film A, the ages include teenagers and adults with a median of 38.